

where,  $f(x_1) \cdot (1 - e^{-\gamma \cdot 1})$  represents the reduction of temporary market impact from the first trade.

Now, if we substitute our first trade price into the equation above we have:

$$P_2 = \{P_0 + f(x_1) + g(x_1) + \varepsilon_1\} + \{f(x_2) + g(x_2)\} - \{f(x_1) \cdot (1 - e^{-\gamma \cdot 1})\} + \varepsilon_2$$

This reduces to:

$$P_2 = P_0 + \underbrace{\{f(x_2) + f(x_1) \cdot e^{-\gamma \cdot 1}\}}_{\text{Cumulative Temporary}} + \underbrace{\{g(x_1) + g(x_2)\}}_{\text{Cumulative Permanent}} + \underbrace{\{\varepsilon_1 + \varepsilon_2\}}_{\text{Cumulative Noise}}$$

where  $f(x_1) \cdot e^{-\gamma \cdot 1}$  is the remaining temporary impact from the first trade.

Following, this formulation, the price in the third period is:

$$P_3 = P_0 + \underbrace{\{f(x_3) + f(x_2) \cdot e^{-\gamma \cdot 1} + f(x_1) \cdot e^{-\gamma \cdot 2}\}}_{\text{Cumulative Temporary}} + \underbrace{\{g(x_1) + g(x_2) + g(x_3)\}}_{\text{Cumulative Permanent}} + \underbrace{\{\varepsilon_1 + \varepsilon_2 + \varepsilon_3\}}_{\text{Cumulative Noise}}$$

After simplifying, we have:

$$P_3 = P_0 + \sum_{j=1}^3 f(x_j) \cdot e^{-\gamma \cdot (3-j)} + \sum_{j=1}^3 g(x_j) + \sum_{j=1}^3 \varepsilon_j$$

In general, the price in period k is:

$$P_k = P_0 + \sum_{j=1}^k f(x_j) \cdot e^{-\gamma \cdot (k-j)} + \sum_{j=1}^k g(x_j) + \sum_{j=1}^k \varepsilon_j$$

With the addition of price drift  $\Delta P$  into our formulation the equation becomes:

$$P_k = P_0 + \sum_{j=1}^k \Delta P_j + \sum_{j=1}^k f(x_j) \cdot e^{-\gamma \cdot (k-j)} + \sum_{j=1}^k g(x_j) + \sum_{j=1}^k \varepsilon_j$$

To estimate the AC model we need to first define our  $f(x)$  and  $g(x)$  impact functions and corresponding parameters, as well as the dissipation impact rate.

$$f(x) = side \cdot a_1 \cdot x^{a_2}$$

$$g(x) = side \cdot b_1 \cdot x^{b_2}$$

$$decay\ function = e^{-\gamma \cdot t}$$